

# **FIRE PERFORMANCE EVALUATION OF NUCLEAR POWER PLANT STRUCTURES**

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# INTRODUCTION: NUCLEAR POWER PLANT STRUCTURES



**Nuclear Power Plant**

- Nuclear power is a clean source of energy without emissions of greenhouse gases.
- According to International Energy Agency (IEA), 10% of global electricity is generated from nuclear power.
- This contribution from nuclear power can be increased to lessen dependence on fossil fuels that generate greenhouse gases.

- There are increasing concerns among people about nuclear waste, an issue yet to be addressed by policy makers.
- The Three Mile Island, Chernobyl, and Fukushima accidents have alerted citizens of the dangers of nuclear power and radiation releases.



Source: <https://www.technologyreview.com/2025/01/16/1110016/new-nuclear-power/>  
<https://doi.org/10.1016/j.nucengdes.2024.113532>



# INTRODUCTION: FIRE SAFETY IN NUCLEAR POWER PLANT



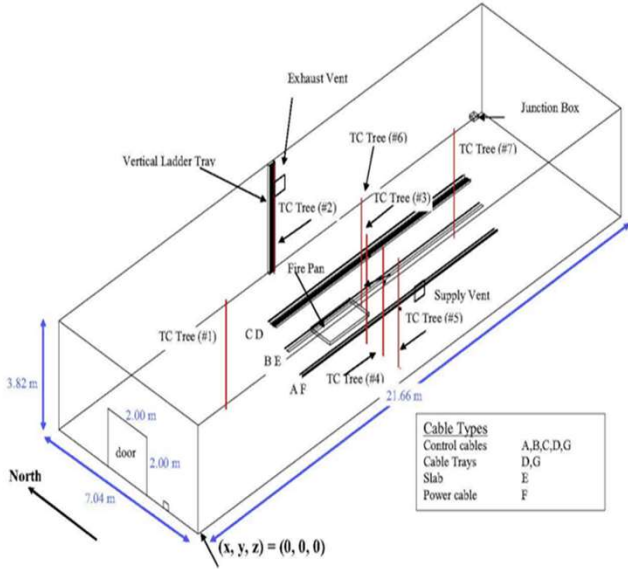
**Browns Ferry Nuclear Plant Aerial View      Browns Ferry Nuclear Plant Fire**

- Prior to Browns Ferry fire in 1975 in United States that almost led to a reactor meltdown, there were minimal regulations to address fire safety of nuclear power stations in US.
- Fire accidents contribute to the risk of nuclear reactor meltdown. Fire incidents can result in failure of redundant cooling systems that keep a reactor cool to prevent a meltdown of the fuel after the nuclear chain reaction is shutdown.
- Performance-based and risk-informed approach are essential for nuclear power plant structures to achieve optimized fire safety and plant performance.



Source: <https://www.nrc.gov/docs/ML1321/ML13210A179.pdf>  
<https://doi.org/10.1016/j.nucengdes.2024.113532>

# LITERATURE REVIEW: FIRE IN NUCLEAR PLANT

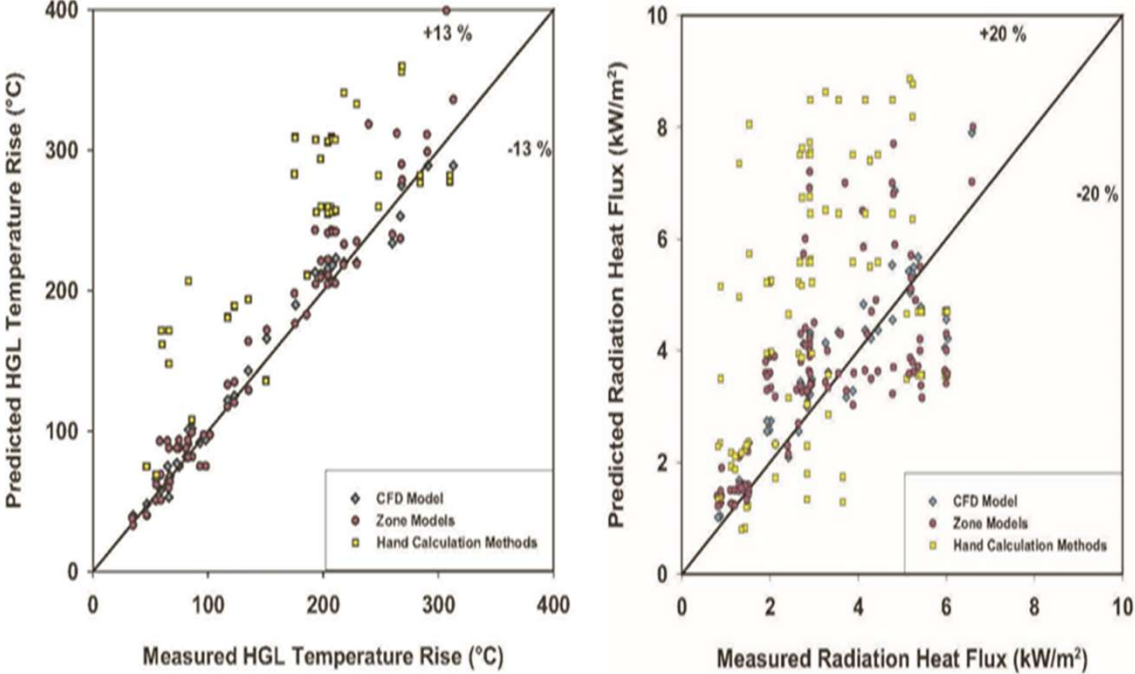
| Author<br>(Year) | Major findings                                                                                                                                                                                                                                                                                                                                                                                                                               | Illustrations                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BRE<br>(2004)    | Conducted a series of full-scale fire experiments at NIST in 2003 to provide data for an international benchmarking exercise to assess the accuracy of fire models over a range of operating conditions expected within a variety of compartments in a nuclear power plant. Seven organizations from France, Germany, Britain, and USA exercised seven fire models as part of the International Collaborative Fire Modeling Project (ICFMP). |  <p data-bbox="890 1092 1257 1198">Schematic of compartment<br/>for Benchmark Exercise<br/>No. 3</p>  <p data-bbox="1507 1089 1856 1198">Hot gas layer in test 3 of<br/>Benchmark Exercise<br/>No.3</p> |



Source: <https://www.govinfo.gov/content/pkg/GOVPUB-C13-27df941f6e9c183edb7e5706decef3a2/pdf/GOVPUB-C13-27df941f6e9c183edb7e5706decef3a2.pdf>  
<https://doi.org/10.1016/j.nucengdes.2024.113532>



# LITERATURE REVIEW: FIRE IN NUCLEAR PLANT

| Author<br>(Year) | Major findings                                                                                                                                                                                                                                                                                                                                           | Illustrations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dey<br>(2024)    | Concluded that fire codes can predict hot gas layer (HGL) temperature, heat flux, gas velocity, etc. reasonably well, provided the heat release rate from fire is given and fire scenario is not too complex. Prediction of flame spread for a given application is found challenging. Future research is required for pyrolysis models for cable fires. |  <p>The figure consists of two side-by-side scatter plots. The left plot shows 'Predicted HGL Temperature Rise (°C)' on the y-axis (0 to 400) versus 'Measured HGL Temperature Rise (°C)' on the x-axis (0 to 400). The right plot shows 'Predicted Radiation Heat Flux (kW/m²)' on the y-axis (0 to 10) versus 'Measured Radiation Heat Flux (kW/m²)' on the x-axis (0 to 10). Both plots include a diagonal line representing perfect agreement. Data points are categorized by model: CFD Model (blue diamonds), Zone Models (red circles), and Hand Calculation Methods (yellow squares). In the temperature plot, most points fall within the -13% to +13% error bands. In the heat flux plot, most points fall within the -20% to +20% error bands.</p> |



Source: <https://www.govinfo.gov/content/pkg/GOVPUB-C13-27df941f6e9c183edb7e5706decef3a2/pdf/GOVPUB-C13-27df941f6e9c183edb7e5706decef3a2.pdf>  
<https://doi.org/10.1016/j.nucengdes.2024.113532>

# RESEARCH GAP

- ❖ Scarcity of realistic three-dimensional models of Nuclear Power Plant Structures.
- ❖ Lack of research on validation of numerical simulation results of Nuclear Power Plant Structures with respect to experimental results.
- ❖ Limited research on the modal performance of Nuclear Power Plant Structures.
- ❖ Limited existing research on the fire performance of Nuclear Power Plant Structures.





# RESEARCH OBJECTIVES & SCOPE

- ❖ To develop three-dimensional models of Nuclear Power Plant Structures.
- ❖ To employ a reasonable model geometry for realistic simulation results of Nuclear Power Plant Structures.
- ❖ To apply suitable material model for capturing nonlinear dynamic response of Nuclear Power Plant Structures.
- ❖ To evaluate the modal performance of Nuclear Power Plant Structures.
- ❖ To assess the fire performance of Nuclear Power Plant Structures (to be completed by next semester).





# METHODOLOGY

A finite element model of the Switchgear room of a Nuclear Power Plant has been developed in ABAQUS software.

The front wall, back wall, side walls, top slab and bottom slab have been created in the Part module of ABAQUS.

M50 concrete material has been defined using the Concrete Damage Plasticity (CDP) model in ABAQUS.

The concrete parts were assembled using “Tie” constraint.

“Encastrate” boundary condition was added to the bottom slab.

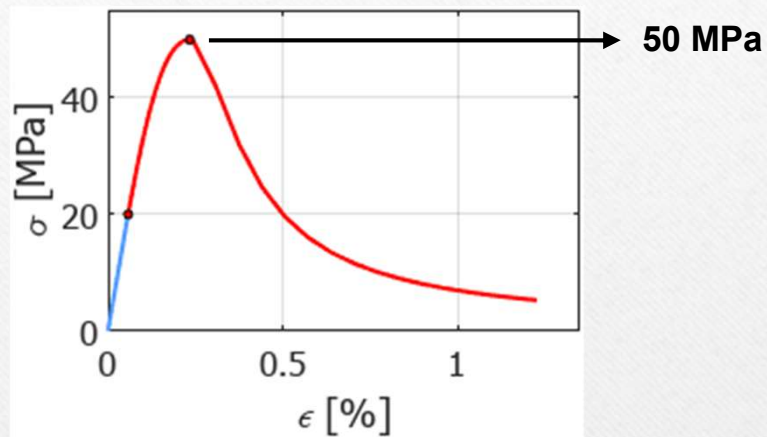
The model was meshed with C3D8R elements of size 0.2 m.

Modal analysis was performed using “Frequency” step.

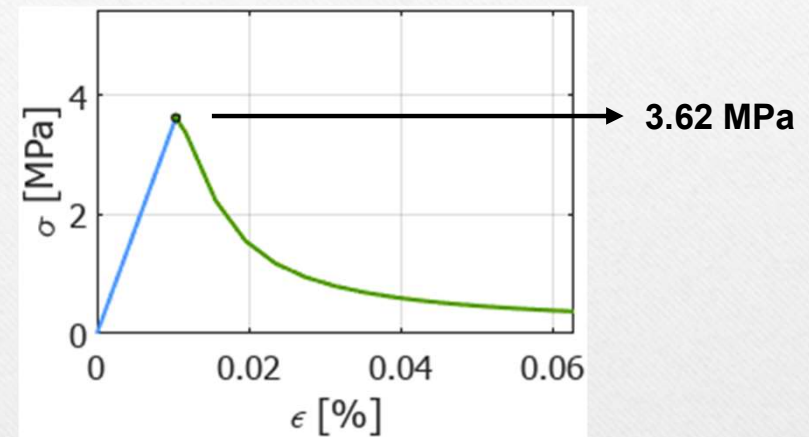


# ABAQUS Material

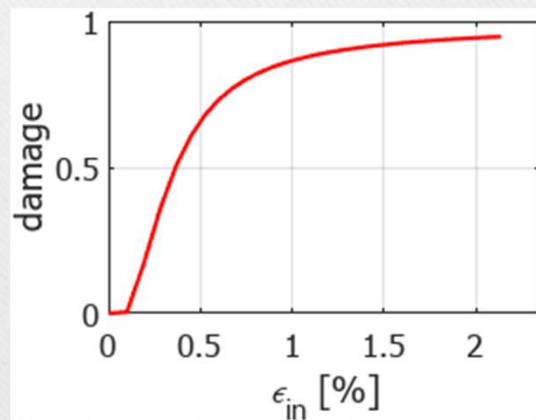
## Concrete Damage Plasticity (CDP) Model – M50 concrete



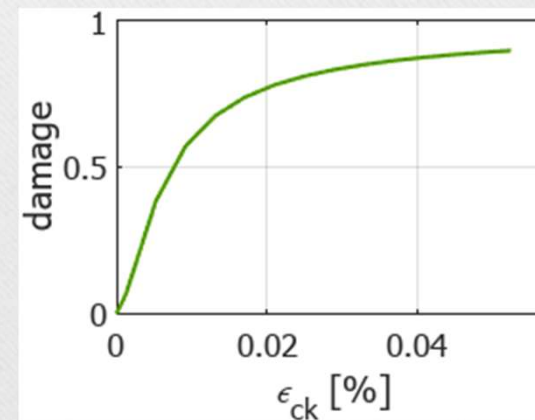
Compressive Stress-Strain Curve



Tensile Stress-Strain Curve



Compressive Damage Curve

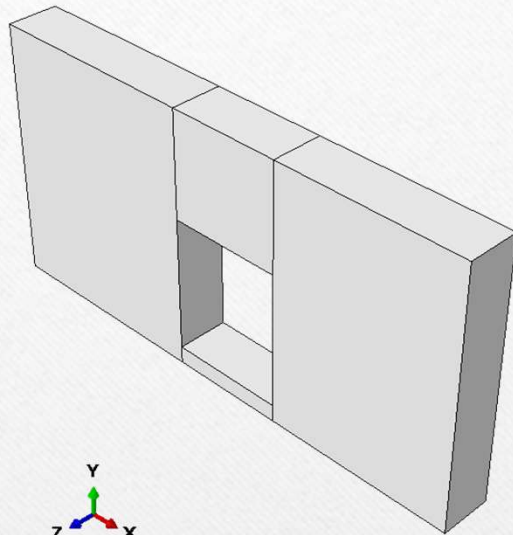


Tensile Damage Curve

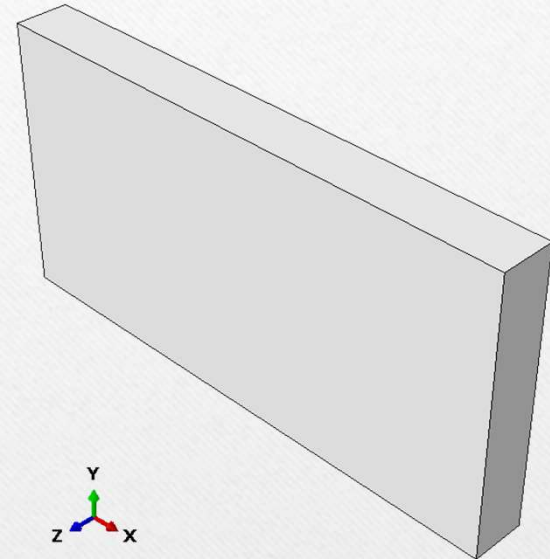


# ABAQUS Parts

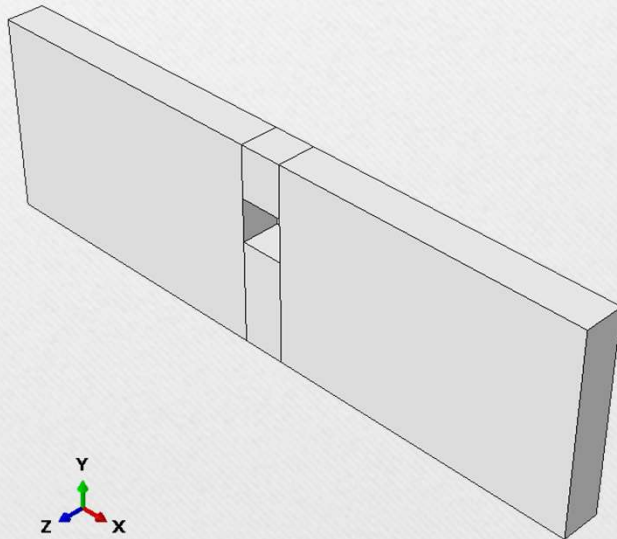
Front wall



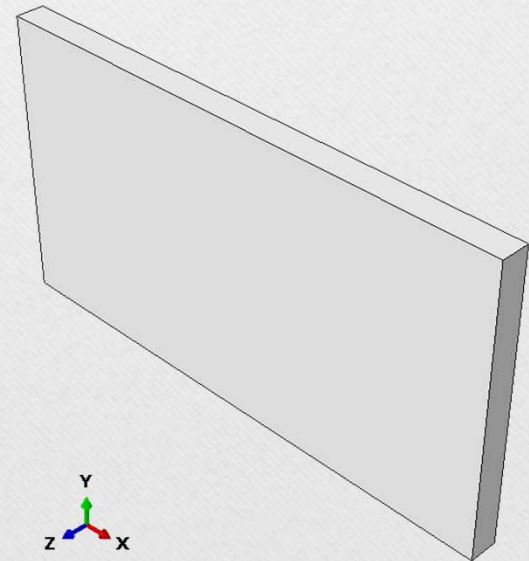
Back wall



Side wall



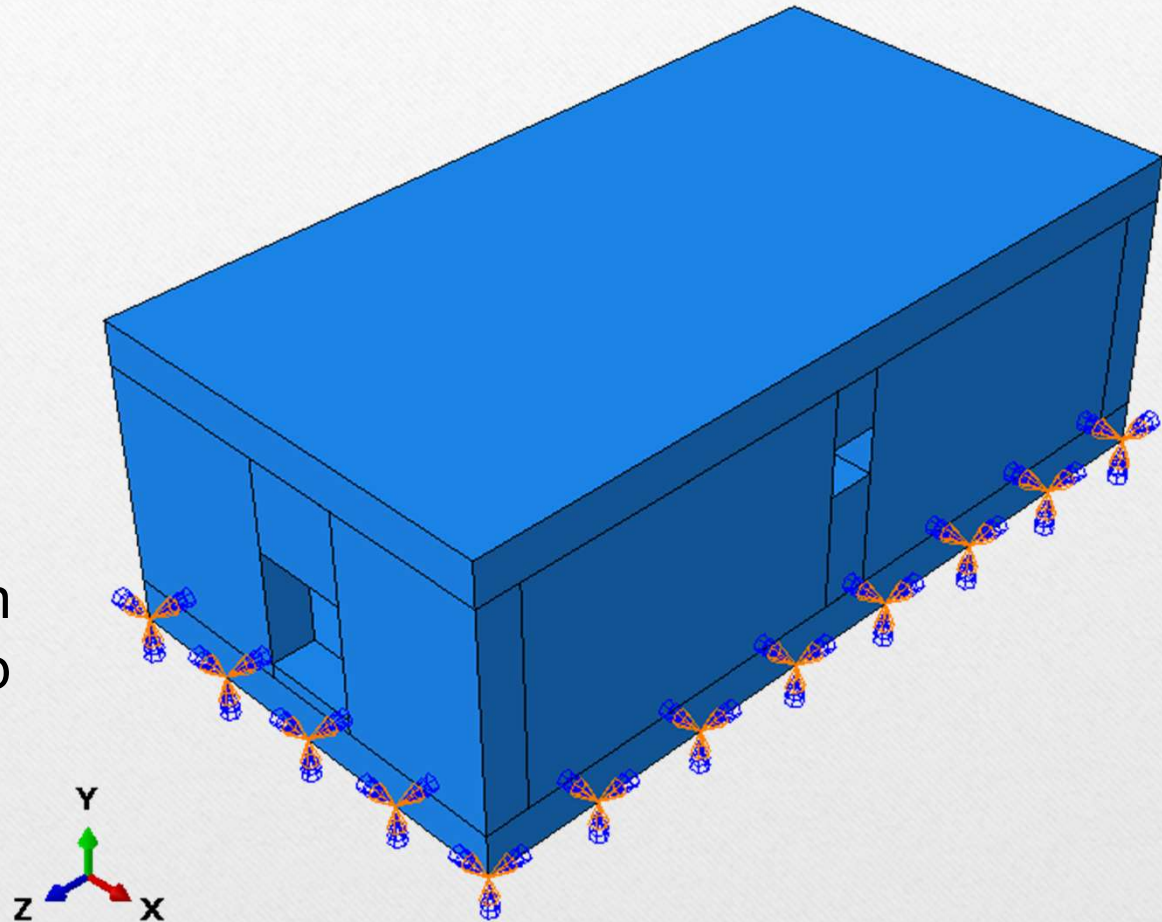
Slab





# ABAQUS Assembly

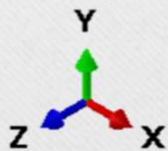
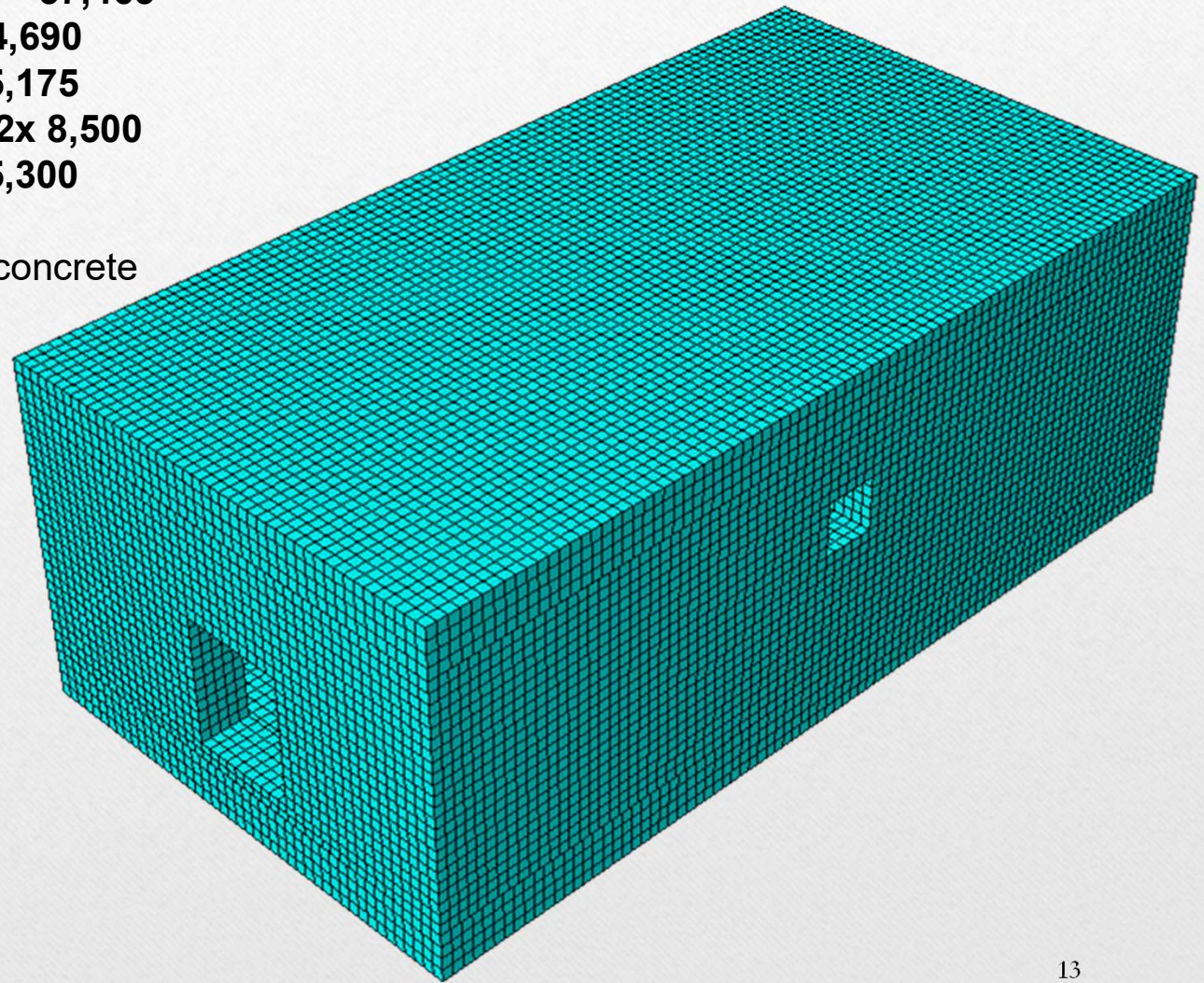
- Concrete Parts connected through **“Tie”** constraint
- **“Encastre”** boundary condition added to base slab





# ABAQUS Meshing

- Total no. of Elements – **57,465**
  - Front Wall – **4,690**
  - Back Wall – **5,175**
  - Side Walls – **2x 8,500**
  - Slabs – **2x 15,300**
- Mesh Size – **0.2 m**
- **C3D8R** elements for concrete

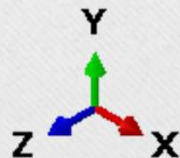
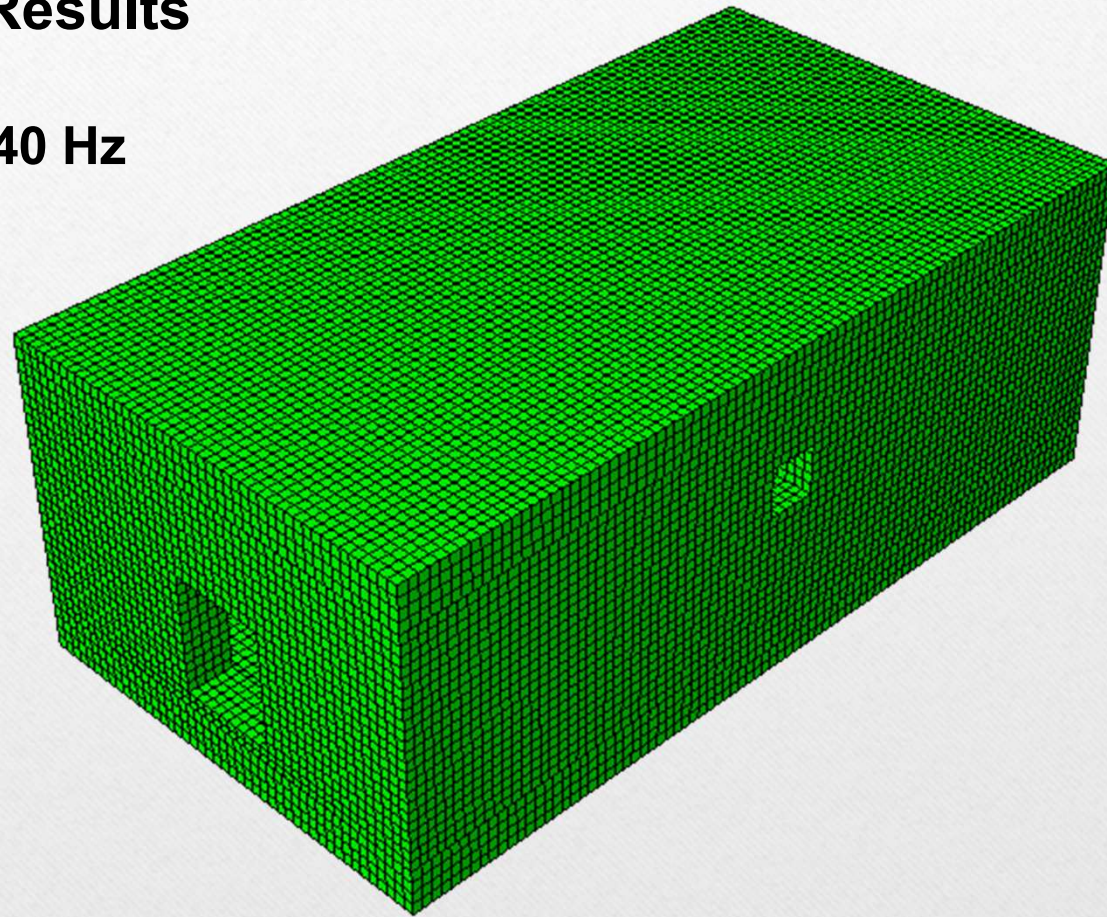




# ABAQUS Results

## Modal Analysis Results

- Mode 1 – 0.76340 Hz



Step: Modal  
Mode

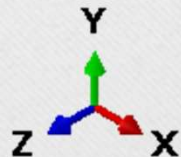
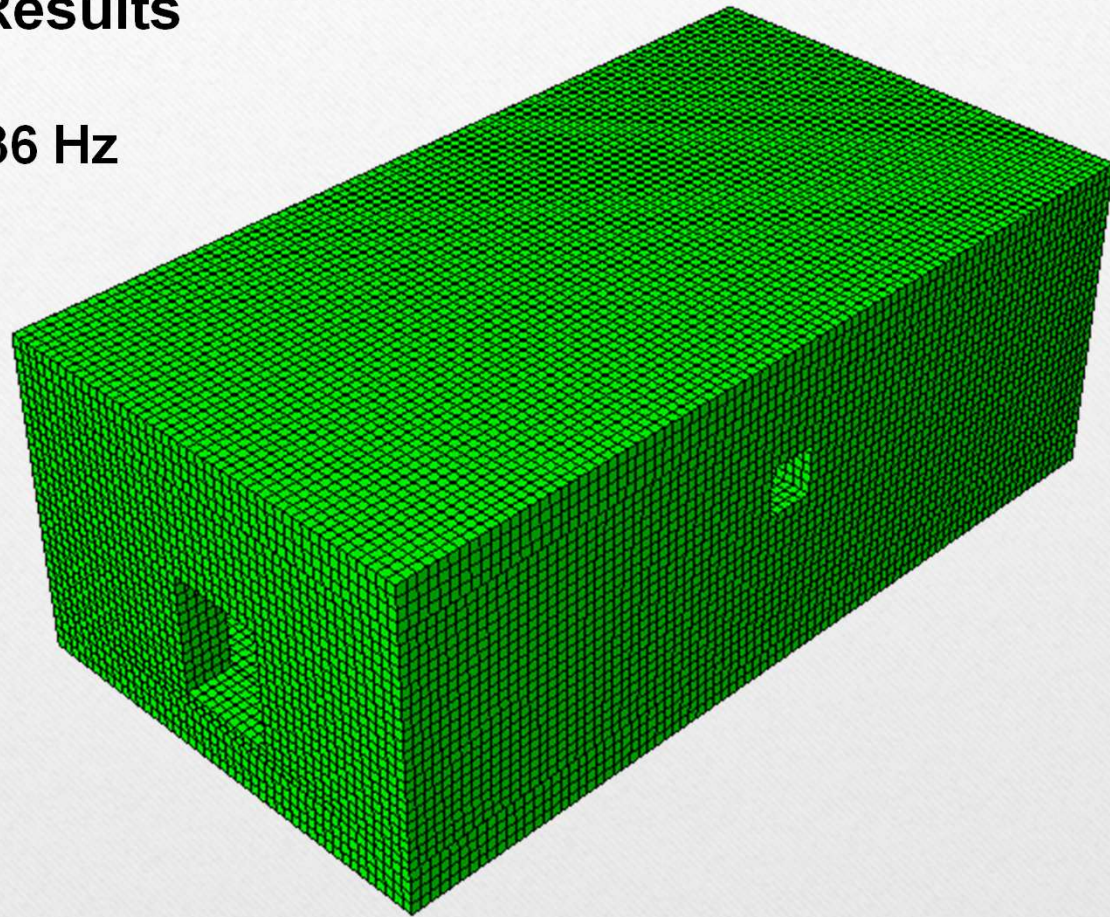
1: Value = 23.007    Freq = 0.76340    (cycles/time)



# ABAQUS Results

## Modal Analysis Results

- Mode 2 – 0.96836 Hz



Step: Modal  
Mode

2: Value = 37.020

Freq = 0.96836

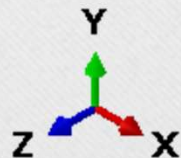
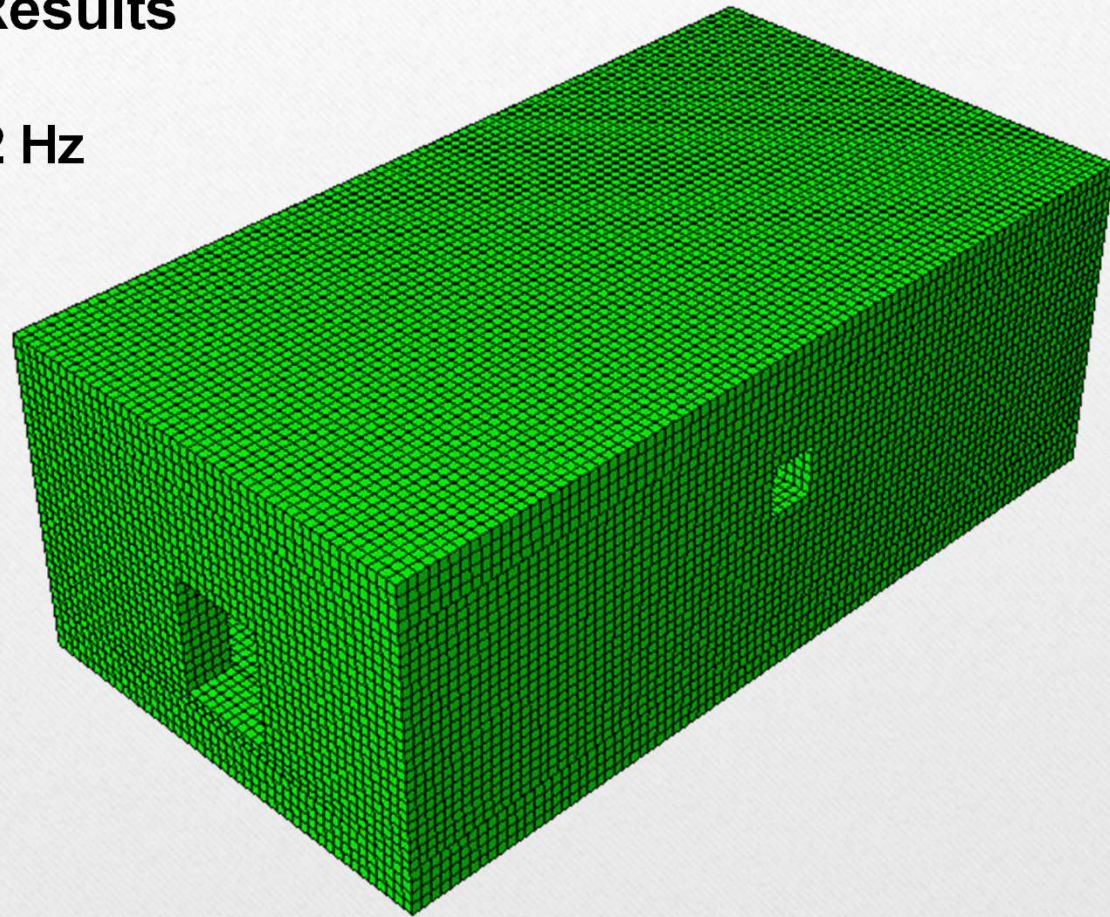
(cycles/time)



# ABAQUS Results

## Modal Analysis Results

- Mode 3 – 1.3102 Hz



Step: Modal  
Mode 3: Value =

67.775

Freq = 1.3102

(cycles/time)



# CONCLUSIONS

- This study presents modal analysis results of the Switchgear room of a Nuclear Power Plant.
- The model geometry is based on research work by Dey *et al.* (2024).
- The model contains C3D8R elements with “Encastré” boundary conditions at the base slab.
- The parts of the model were connected through “Tie” constraints.
- The natural frequency of the structure is 0.7634 cycles/second.
- The fire simulation will be performed in the next semester.





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*Thank You*

